5.12. Consider the AIRLINE relational database schema shown in Figure 5.8, which describes a database for airline flight information. Each FLIGHT is identified by a Flight_number, and consists of one or more FLIGHT_LEGs with Leg_numbers 1, 2, 3, and so on. Each FLIGHT_LEG has scheduled arrival and departure times, airports, and one or more LEG_INSTANCEs— one for each Date on which the flight travels. FAREs are kept for each FLIGHT. For each FLIGHT_LEG instance, SEAT_RESERVATlONs are kept, as are the AIRPLANE used on the leg and the actual arrival and departure times and airports. An AIRPLANE is identified by an Airplane_id and is of a particular AIRPLANE_TYPE. CAN_LAND relates AIRPLANE_TYPEs to the AIRPORTs at which they can land. An AIRPORT is identified by an Airport_code. Consider an update for the AIRLINE database to introduce a new leg instance tuple in the LEG_INSTANCE relation.

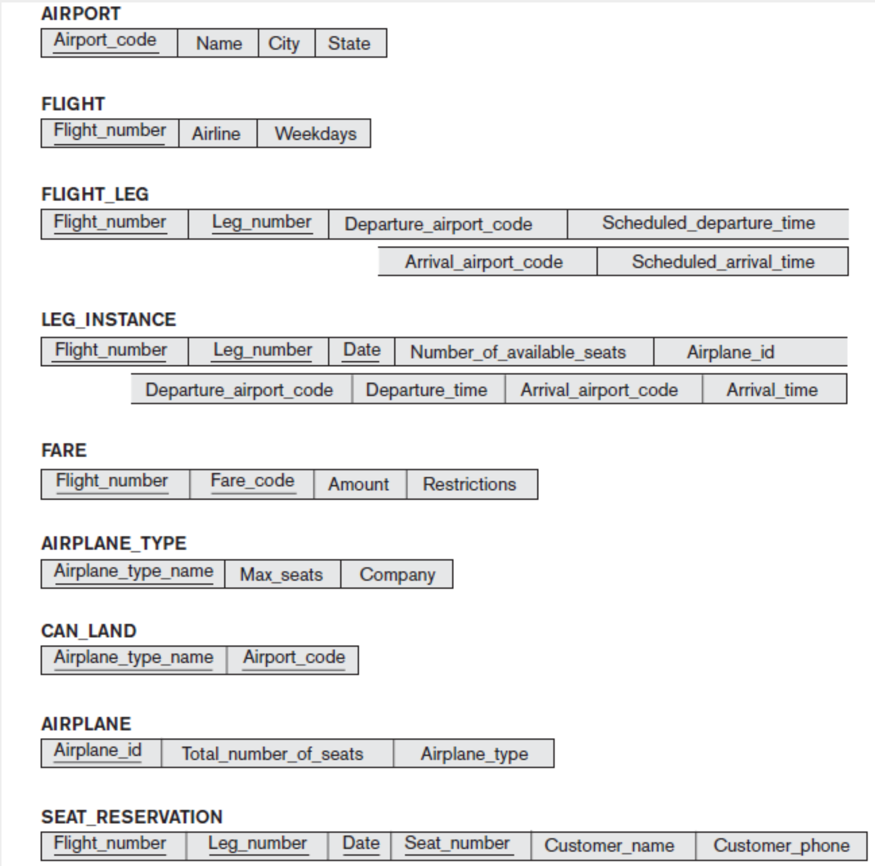


Figure 5.8
The AIRLINE relational database schema.

a. Give the operations for this update.

b. What types of constraints would you expect to check?

c. Which integrity constraints should be checked to insert a new flight leg tuple into the FLIGHT_LEG relation not?

d. Specify all the referential integrity constraints that hold on the schema shown in Figure 5.8.

a. 假設小駝峰寫法為對應欄位的數值

- INSERT <flightNumber, legNumber, date, numberOfAvailableSeats, airplaneId, departureAirportCode, departureTime, arrivalAirportCode, arrivalTime> into LEG_INSTANCE

b.

- Entity integrity constraints：必須要有 Flight_number 和 Leg_number
- Referential Integrity constraints：因為 FLIGHT_LEG 的 Flight_number 和 Leg_number 為 PK，一定要有同樣 Flight_number 和 Leg_number 的 FLIGHT_LEG 存在
- Semantic constraints：Departure_airport_code 和 Arrival_airport_code 要和 FLIGHT_LEG 一致才合理

c. Referential Integrity constraints

d.

Referenced		Referencing	
AIRPOTY	Airport_code	FLIGHT_LEG	Departure_airport_code
		FLIGHT_LEG	Arrival_airport_code
		LEG_INSTANCE	Departure_airport_code
		LEG_INSTANCE	Arrival_airport_code
		CAN_LAND	Airport_code
FLIGHT	Flight_number	FLIGHT_LEG	Flight_number
		LEG_INSTANCE	Flight_number
		FARE	Flight_number
		SEAT_RESERVATION	Flight_number
FLIGHT_LEG	Leg_number	LEG_INSTANCE	Leg_number
		SEAT_RESERVATION	Leg_number
LEG_INSTANCE	Date	SEAT_RESERVATION	Date
AIRPLANE_TYPE	Airplane_type_name	CAN_LAND	Airplane_type_name
AIRPLAINE	Airplaine_id	LEG_INSTANCE	Airplaine_id

e.

5.17. Consider the following relations for a database that keeps track of booking of apartments by a constructor. (OPTION refers to some specific optional requirements/designs stated by the client to be implemented in the flat):

APARTMENT(Apartment#, Model, Address, Price_perSquareFt)

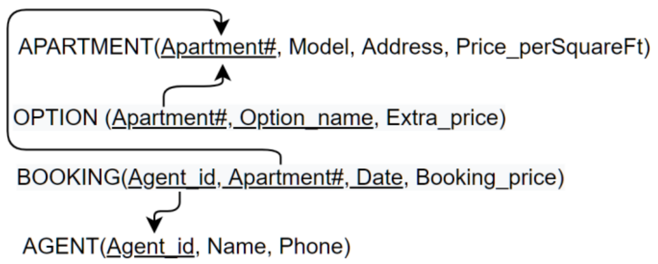
OPTION (Apartment#, Option_name, Extra_price)

BOOKING(Agent_id, Apartment#, Date, Booking_price)

AGENT(Agent_id, Name, Phone)

First, specify the foreign keys for this schema, stating any assumptions you make. Next, populate the relations with a few sample tuples, and then give an example of an insertion in the BOOKING and AGENT relations that violates the referential integrity constraints and of another insertion that does not.

1. 假設 Schema 為下圖：



FK 為 BOOKING 的 Agent_id、Apartment#和 OPTION 的 Apartment#。

2. APARTMENT

Apartment#	Model	Address	Price_perSquareFt
01	單人房	基隆路1號	800
02	雙人房	基隆路2號	800

OPTION

Apartment#	Option_name	Extra_price
01	加床	800

BOOKING

Agent_id	Apartment#	Date	Booking_price
001	01	2020/5/5	4000
002	02	2020/5/7	5600

AGENT

Agent_id	Name	Phone

001	John	0900123456
002	Mark	0900654321

INSERT BOOKING 要破壞 referential integrity constraints 的話，例如加入 <003,03,2020/5/5,8000>；不破壞的話例如加入<001,02,2020/5/5,8000>。

INSERT AGENT 不會破壞 referential integrity constraints。

8.15. Show the result of each of the sample queries in Section 8.5 as it would apply to the database state in Figure 5.6.

Figure 5.6
One possible database state for the COMPANY relational database schema.

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

Query 1. Retrieve the name and address of all employees who work for the ‘Research’ department.

Fname	Minit	Lname	Address
John	B	Smith	731 Fondren, Houston, TX

Franklin	T	Wong	638 Voss, Houston, TX
Ramesh	K	Narayan	975 Fire Oak, Humble, TX
Joyce	A	English	5631 Rice, Houston, TX

Query 2. For every project located in ‘Stafford’, list the project number, the controlling department number, and the department manager‘s last name, address, and birth date.

Pnumber	Dnum	Lname	Address	Bdate
10	4	Wallace	291 Berry, Bellaire, TX	1941-06-20
30	4	Wallace	291 Berry, Bellaire, TX	1941-06-20

Query 3. Find the names of employees who work on all the projects controlled by department number 5.

Fname	Minit	Lname
John	B	Smith
Joyce	A	English
Franklin	T	Wong
Ramesh	K	Narayan

Query 4. Make a list of project numbers for projects that involve an employee whose last name is ‘Smith’, either as a worker or as a manager of the department that controls the project.

Pno
1
2

Query 5. List the names of all employees with two or more dependents. Strictly speaking, this query cannot be done in the basic (original) relational algebra. We have to use the AGGREGATE FUNCTION operation with the COUNT aggregate function. We assume that dependents of the same employee have distinct Dependent_name values.

Fname	Minit	Lname
John	B	Smith
Franklin	T	Wong

Query 6. Retrieve the names of employees who have no dependents. This is an example of the type of query that uses the MINUS (SET DIFFERENCE) operation.

Fname	Minit	Lname
Alicia	J	Zelaya
Ramesh	K	Narayan
Joyce	A	English
Ahmad	V	Jabbar
James	E	Borg

Query 7. List the names of managers who have at least one dependent.

Fname	Minit	Lname
Franklin	T	Wong
Jennifer	S	Wallance

8.17. Consider the AIRLINE relational database schema shown in Figure 5.8, which was described in Exercise 5.12. Specify the following queries in relational algebra:

a. For each flight, list the flight number, airline, date, departure airport and arrival airport along with the number of available seats, for each leg.

b. For all the airplane types available in the database, list the airplane type, name of the company, maximum seats available and the airports (or airport codes) where these planes can land.

c. List the name, seat number and phone numbers of all the customers of all flights or flight legs that departed from Houston Intercontinental Airport (airport code 'iah') and arrived in Los Angeles International Airport (airport code 'lax') on '2016-03-16'.

d. List fare information for all the flights run by the airline 'Delta Airlines'.

e. Retrieve the number of available seats on all flights run by Delta Airline, on '2016-04-09'.

a. $ALL\_LEG \leftarrow FLIGHT \bowtie_{Flight\_number=Flight\_number} LEG\_INSTANCE$

$RESULT \leftarrow$

$\pi_{Flight\_number,Airline,Date,Departure\_airport\_code,Arrival\_airport\_code,Number\_of\_available\_seats}(ALL\_LEG)$

b. $RESULT \leftarrow AIRPLAINE\_TYPE \bowtie_{Airplain\_type\_name=Airplain\_type\_name} CAN\_LAND$

c. $LEGS \leftarrow$

$\sigma_{Departure\_airport\_code='iah'\ AND\ Arrival\_airport\_code='lax'\ AND\ Date='2016-03-16'}(LEG\_INSTANCE)$

$RESULT \leftarrow \pi_{Customer\_name,Seat\_number,Customer\_phone}(LEGS$

$\bowtie_{Flight\_number=Flight\_number\ AND\ Leg\_number=Leg\_number\ AND\ Date=Date} SEAT\_RESERVATION)$

d. $RESULT \leftarrow \pi_{Flight\_number}(\sigma_{Airline='Delta\ Airlines'}(FLIGHT)) \bowtie_{Flight\_number=Flight\_number} FARE$

e. $RESULT \leftarrow \pi_{Number\_of\_available\_seats}(\sigma_{Date='2016-04-09'}$

$(\pi_{Flight\_number}(\sigma_{Airline='Delta\ Airlines'}(FLIGHT)) \bowtie_{Flight\_number=Flight\_number} LEG\_INSTANCE))$

8.22. Consider the two tables T1 and T2 shown in Figure 8.15. Show the results the following operations:

a. $T1 \bowtie_{T1.P=T2.A} T2$

b. $T1 \bowtie_{T1.Q=T2.B} T2$

c. $T1 ⟕_{T1.P=T2.A} T2$

d. $T1 ⟖_{T1.Q=T2.B} T2$

e. $T1 \cup T2$

Figure 8.15
A database state for the relations *T1* and *T2*.

TABLE T1

P	Q	R
10	a	5
15	b	8
25	a	6

TABLE T2

A	B	C
10	b	6
25	c	3
10	b	5

a.

P	Q	R	A	B	C
10	a	5	10	b	6
10	a	5	10	b	5
25	a	6	25	c	3

b.

P	Q	R	A	B	C
15	b	8	10	b	6
15	b	8	10	b	5

c.

P	Q	R	A	B	C
10	a	5	10	b	6

10	a	5	10	b	5
15	b	8	null	null	null
25	a	6	25	c	3

d.

P	Q	R	A	B	C
15	b	8	10	b	6
null	null	null	25	c	3
15	b	8	10	b	5

e.

P	Q	R
10	a	5
15	b	8
25	a	6
10	b	6
25	c	3
10	b	5

8.25. Specify queries a, b, c, and d of Exercise 8.17 in both tuple and domain relational calculus.

a. For each flight, list the flight number, airline, date, departure airport and arrival airport along with the number of available seats, for each leg.

b. For all the airplane types available in the database, list the airplane type, name of the company, maximum seats available and the airports (or airport codes) where these planes can land.

c. List the name, seat number and phone numbers of all the customers of all flights or flight legs that departed from Houston Intercontinental Airport (airport code 'iah') and arrived in Los Angeles International Airport (airport code 'lax') on '2016-03-16'.

d. List fare information for all the flights run by the airline 'Delta Airlines'.

a. Tuple: $\{f.Flight\_number,\ f.Airline, l.Date,\ l.Departure\_airport\_code,\ l.Arrival\_airport\_code,$ $l.Number\_of\_available\_seats \mid FLIGHT(f)\ and\ (\exists l)(LEG\_INSTANCE(l)\ and\ f.Flight\_number =$

$l.Flight\_number)\}$

Domain: $\{abfikg \mid (\exists d)(FLIGHT(abc)\ and\ LEG\_INSTANCE(defghijkl)\ and\ a = d)\}$

b. Tuple: $\{t.Airplaine\_type\_name, t.Max\_seats, t.Company,$

$c.Airport\_code \mid AIRPLAINE\_TYPE(t)\ and\ (\exists c)(CAN\_LAND(c)\ and\ t.Airplaine\_type\_name =$
$c.Airplaine\_type\_name)\}$

Domain: $\{abce \mid (\exists d)(AIRPLAINE\_TYPE(abc)\ and\ CAN\_LAND(de)\ and\ c = d)\}$

c. Tuple: $\{s.Customer_{name}, s.Seat_{number}, s.Customer_{phone} | SEAT_{RESERVATION(s)}$

$and\ (\exists l)(LEG\_INSTANCE(l)\ and\ s.Flight\_number = l.Flight\_number\ and\ s.Leg\_number$
$= l.Leg\_number\ and\ s.Date = l.Date\ and\ l.Department\_airport\_code$
$= 'iah'\ and\ l.Arrival\_airport\_code = 'lax'\ and\ l.Date = '2016-03-16')\}$

Domain: $\{mno \mid$

$(\exists j)(\exists k)(\exists l)(LEG\_INSTANCE(abcdefghi)\ and\ SEAT\_RESERVATION(jklmno)\ and\ a = j\ and\ b$
$= k\ andc = l)\}$

d. Tuple: $\{f.Flight\_number,\ f.Fare\_code,\ f.Amount,$

$f.Restrictions \mid FARE(f)\ \ and\ (\exists l)(FLIGHT(t)\ and\ t.Airline = 'Delta\ Airlines'\ and\ f.Flight\_number =$
$t.Flight\_number)\}$

Domain: $\{abcd \mid (\exists e)(\exists f)(FARE(abcd)\ and\ FLIGHT(efg)\ and\ a = e\ and\ f = 'Delta\ Airlines')\}$

8.30. Show how you can specify the following relational algebra operations in both tuple and domain relational calculus.

a. $\sigma_{A=C}(R(A,B,C))$

c. $\big(R(A,B,C)\big) * S(C,D,E)$

d. $\big(R(A,B,C)\big) \cup S(A,B,C)$

f. $\big(R(A,B,C)\big) = S(A,B,C)$

g. $\big(R(A,B,C)\big) \times S(A,B,C)$

a. Tuple: $\{r \mid R(r)\ and\ r.A = r.C\}$

Domain: $\{abc \mid (\exists c)(R(abc)\ and\ a = c)\}$

c. Tuple: $\{r.A, r.B, r.C, s.D, s.E | R(r)\ and\ S(s)\ and\ r.C = s.C\}$

Domain: $\{abcef \mid (\exists d)(R(abc)\ and\ S(def)\ and\ c = d)\}$

d. Tuple: $\{t \mid R(t)\ or\ S(t)\}$

Domain: $\{abc \mid R(abc)\ or\ S(abc)\}$

f. Tuple: $\{r \mid R(r)\ and\ S(s)\ and\ (r.A \neq s.A\ or\ r.B \neq s.B\ or\ r.C \neq s.C)\}$

Domain: $\{abc \mid (\exists d)(\exists e)(\exists f)(R(abc)\ and\ S(def)\ and\ (a \neq d\ or\ b \neq e\ or\ c \neq f))\}$

g. Tuple: $\{r.A, r.B, r.C, s.A, s.B, s.C \mid R(r) \text{ and } S(s)\}$

Domain: $\{abcdef \mid R(abc) \text{ and } S(def)\}$

9.4. Figure 9.8 shows an ER schema for a database that can be used to keep track of transport ships and their locations for maritime authorities. Map this schema into a relational schema and specify all primary keys and foreign keys.

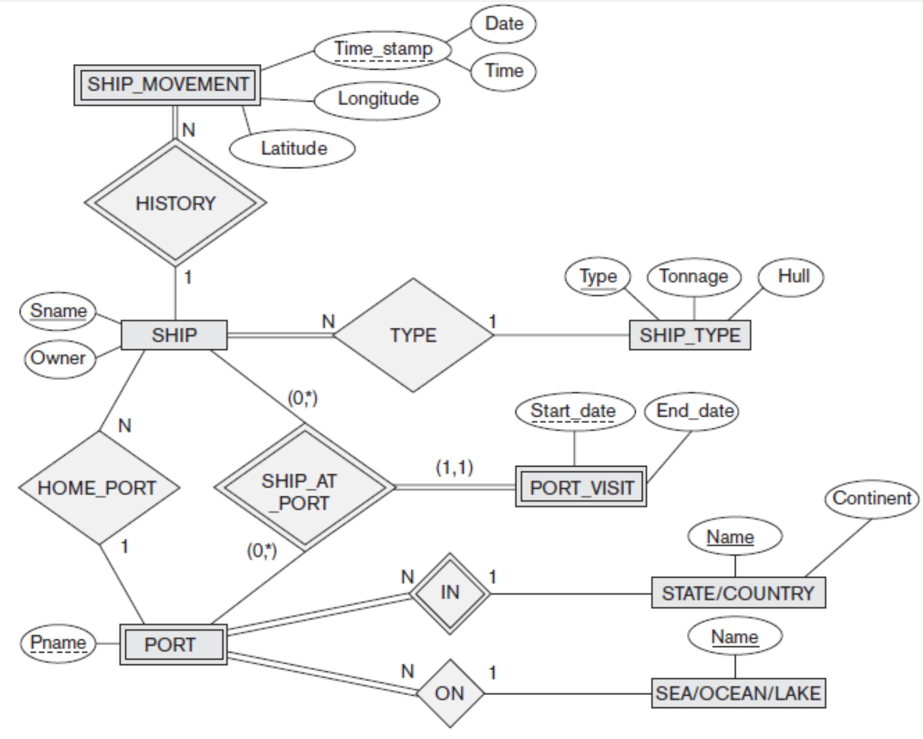


Figure 9.8
An ER schema for a SHIP_TRACKING database.